

JEE Main & Advanced 15-Year Solved Physics Archive

Target Exam: JEE Main & JEE Advanced 2026 | Stream: Engineering | Subject: Physics

Verified Official Academic Preparation Material - TheCampusNova Learning Portal

Key Topics Covered: Electrodynamics, Mechanics & Modern Physics

1. Mechanics: Conservation of Linear Momentum, Rotational Dynamics ($I = \sum(m \cdot r^2)$), Torque = $I \cdot \alpha$
2. Electrodynamics: Coulomb Law ($F = q_1 \cdot q_2 / 4 \cdot \pi \cdot \epsilon_0 \cdot r^2$), Gauss Law (Flux = Q_{in} / ϵ_0)
3. Biot-Savart Law: $dB = (\mu_0 / 4 \cdot \pi) \cdot (I \cdot dl \times r) / r^3$ | Ampere Circuital Law: $\int(B \cdot dl) = \mu_0 \cdot I_{enc}$
4. Modern Physics: Photoelectric Equation ($E_k = h \cdot \nu - \Phi$), de Broglie wavelength ($\lambda = h/p$)
5. Solved PYQ 1: Find magnetic field at center of circular loop carrying current I . Answer: $B = \mu_0 \cdot I / (2 \cdot R)$
6. Solved PYQ 2: A particle undergoes SHM with amplitude A . At what displacement is $KE = PE$? Answer: $x = A / \sqrt{2}$
7. Exam Strategy: Solve Mechanics first only if calculation speed is high; Modern Physics gives guaranteed marks with zero risk.